



Thermal shock behavior of 3-dimensional C/SiC composite

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Abstract

The thermal shock behavior of a three-dimensional carbon fiber reinforced SiC matrix fabricated by chemical vapor infiltration (CVI) technique was studied using the air quenched method. Damage to composites was assessed by a destructive technique of measuring mechanical properties using three-point flexure and SEM characterization. C/SiC composites displayed good resistance to thermal shock, and retained 83% of the original strength after quenching from 1300 to 300°C 100 times. The critical ΔT of C/SiC in combustion environment was 700°C. The critical number of thermal shocks for the C/SiC composite was about 50 times. When the number of thermal shocks was less than 50 times, the residual flexural strength of C/SiC composites decreased with the increase of thermal shock times. When the number of thermal shocks of C/SiC was greater than 50, the strength of C/SiC did not further decrease because the crack density was saturated. © 2002 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Carbon-fiber-reinforced SiC-matrix composite fabricated by the chemical vapor infiltration (CVI) process (C/SiC-CVI) has been developed for potential use in gas-turbine engines [1,2]. C/SiC composite is expected to be suitable for high-temperature use, but thermal shock and thermal cycling effects are anticipated to be one factor limiting performance in many instances, so that thermal shock damage must be understood prior to use.

Thermal shock resistance of monolithic materials has been extensively studied, and theoretical analyses have been successfully applied to explain experimental observations [3–6]. However, a comprehensive understanding of the thermal shock behavior of carbon-fiber-reinforced ceramic composites (CFCCs) has not been obtained, despite recent advances in the architecture design and processing of these materials. Experimental thermal shock

studies have been conducted on unidirectional, 0–90°, and two-dimensional woven-fiber composites [7–11]. These studies have shown that thermal shock damage results in noncatastrophic strength decreases above a critical quench temperature difference, ΔT_c , in contrast to monolithic ceramics that typically exhibit a catastrophic decrease of strength at ΔT_c . Furthermore, these composites retain a significant portion of their prequench strength above ΔT_c , whereas monolithic ceramics are completely fractured or can only support an insignificant load.

Kagawa et al. studied the thermal shock resistance of uniaxial-SiC-fiber composites with borosilicate glass and lithium aluminosilicate (LAS) matrices [12]. The borosilicate composite exhibited decreases in modulus and flexural strength at $\Delta T > 600^\circ\text{C}$, whereas the LAS composite showed no degradation in mechanical properties at ΔT up to 1000°C. Singh et al. studied the thermal shock resistance of the two-dimensional Nicalon fiber reinforced SiC matrix fabricated by chemical vapor infiltration (CVI) technique [13,14]. The SiC/SiC composites exhibited decreases in flexural strength at $\Delta T > 700^\circ\text{C}$. The thermal shock damage in all of these composites consisted of matrix cracks.

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Up to now, our knowledge on the behavior of carbon fiber-reinforced SiC matrix composite subjected to thermal transients is rather limited. The thermal shock behavior of C/SiC(CVI) is tested in the present study.

2. Experimental procedure

2.1. Preparation of C/SiC composite

T-300™ carbon fiber from Toray (Japan) was employed. The fiber preform was prepared using a three-dimensional braid method, and was supplied by the Nanjing Institute of Glass Fiber, People's Republic of China. The volume fraction of fibers was about 40%. LCVI was employed to deposit a pyrolytic carbon layer and the silicon carbide matrix. A thin pyrolytic carbon layer was deposited on the surface of the carbon fiber as the interfacial layer with C_3H_8 at 800°C. Methyltrichlorosilane (MTS, CH_3SiCl_3) was used for the deposition of the SiC matrix. MTS vapor was carried by bubbling hydrogen. Typical conditions for deposition were 1000°C, a hydrogen:MTS ratio of 10, and a pressure of 5 kPa. Argon was employed as the dilute gas to slow down the chemical reaction rate of deposition. The dimension of as-received C/SiC sample was 4 mm×6 mm×140 mm. In order to improve the thermal stability of the mechanical properties, the as-received samples were heat-treated in a vacuum at 1300°C for 3 h.

2.2. Thermal shock tests

Thermal shock tests were conducted in burner rigs. A schematic of the burner rig system is illustrated in Fig. 1. The combustion temperature was measured by temperature probe along the length direction of test samples from the flame center. The samples were kept in the combustion atmosphere for 30 s and then cooled in an air atmosphere for 60 s. The thermal shock temperature difference parallel

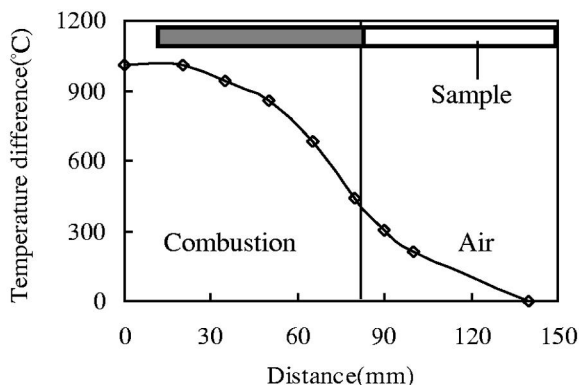


Fig. 2. Relation curve between temperature difference and distance from flame center.

to the length direction of the sample was from 0 to 1000°C, as shown in Fig. 2.

2.3. Measurement of composites

Flexural strength was measured using the three-point-bending method. The span dimension was 20 mm, and the loading rate was 0.5 mm min⁻¹. The tests were conducted on an Instron-1195 device at room temperature. The fracture surface was observed with a scanning electron microscope (SEM, JEOL JXA-840). The microstructure of the texture was analyzed with a transmission electron microscope (TEM, JEOL FX-2000).

3. Results and discussion

3.1. Effect of ΔT on the mechanical properties

The assessment of thermal shock damage was done by comparing load-displacement curves of unquenched and

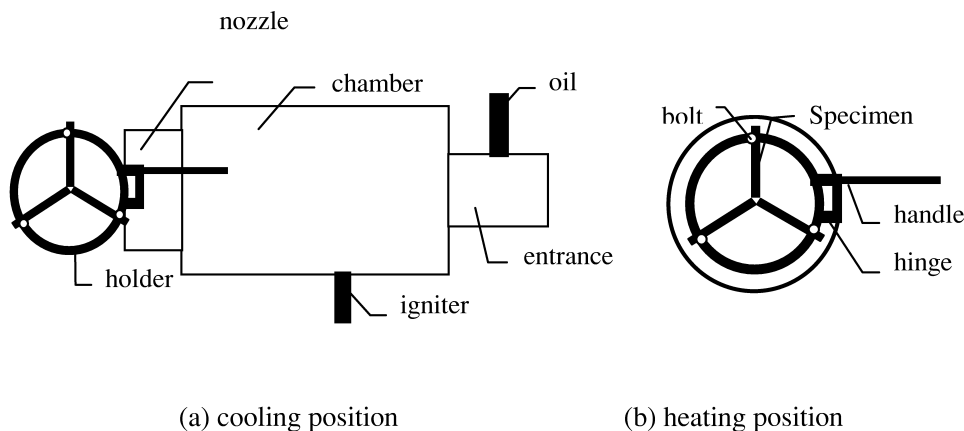


Fig. 1. Schematic of burner rig and thermal-shock device.

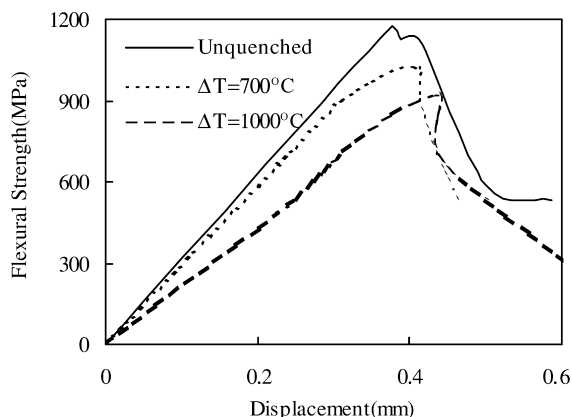


Fig. 3. Influence of thermal shock on load-displacement behavior of C/SiC composites.

quenched composites. The influence of thermal shock on the load-displacement behavior of C/SiC composite is shown in Fig. 3. It is apparent from these curves that the mechanical properties of thermally shocked composites are different from the unquenched sample. Thermally shocked samples show changes in a number of composite characteristics. The elastic modulus and matrix cracking strength are lower for quenched composites because of the lower slope of the linear portion of the curve and earlier deviation from linearity, respectively. In addition, the ultimate strength, and work-of-fracture (area under the curve) are also lower for composites after thermal shock than the as-received samples. Similar load-displacement curves were obtained for Nicalon/CVI SiC and Nicalon/polymer SiC composites [14].

The effects of thermal shock on the maximum flexural strength may result from the following two causes: the oxidation of fibers, or the fracture of fibers under the thermal stress. The decrease of modulus of composites may be due to matrix cracking, fiber fracture and interface debonding.

When C/SiC composites were cooled from high temperature to low temperature, the tension stress on the surface and the compressive stress in the interior are created by the thermal gradient, which causes the matrix cracks on the surface of composites. High strength fibers hindered the propagation of matrix cracks, and the mechanical degradation was reduced by crack deflection and crack bridging. Therefore, the bonding between fiber and matrix should be weak enough in order to make the fiber and matrix debond and the cracks deflect around fibers. Otherwise, the matrix cracks will propagate into the fibers and reduce the strength of the composites.

A C-interface exists between the C-fibers and SiC matrix, with a thickness of about 0.3 μm (Fig. 4), which ensures the weak bonding between C fibers and SiC matrix. Therefore, C/SiC composites show tough fracture

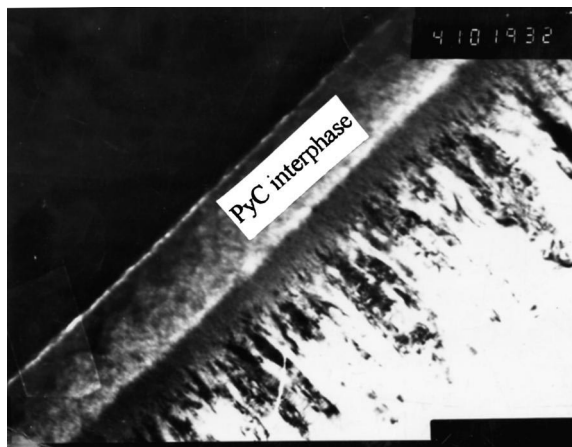


Fig. 4. TEM micrograph of PyC interphase between fiber and matrix.

characteristics. As shown in Fig. 5, C/SiC shows apparent characteristics of fiber pull-out after being thermally shocked 100 times at $\Delta T=700^\circ\text{C}$.

As shown in Fig. 6, at $\Delta T < 700^\circ\text{C}$, the flexural strength of C/SiC does not decrease with the increase of ΔT . When ΔT is larger than 700°C , the flexural strength decreases with the increase of ΔT . As can be shown, the critical thermal shock temperature difference (ΔT_c) is 700°C .

According to the thermoelastic theories, when a material is subjected to a thermal transient of suddenly decreasing temperature (ΔT), e.g. in a quench test, the surface of the material is placed under a tensile stress and the interior under a compressive stress. When the temperature difference equals a critical value (ΔT_c), the tensile thermal stress generated is sufficient to cause the formation of a surface crack. The tensile strength (σ_t) can be expressed as follows [14]:

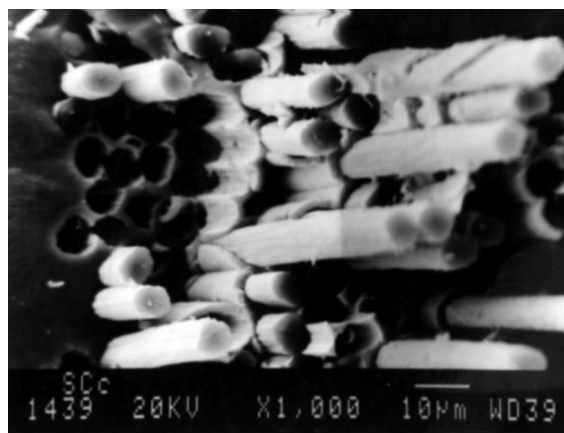


Fig. 5. Morphology of fracture section of C/SiC after thermal-shock ($\Delta T=700^\circ\text{C}$).

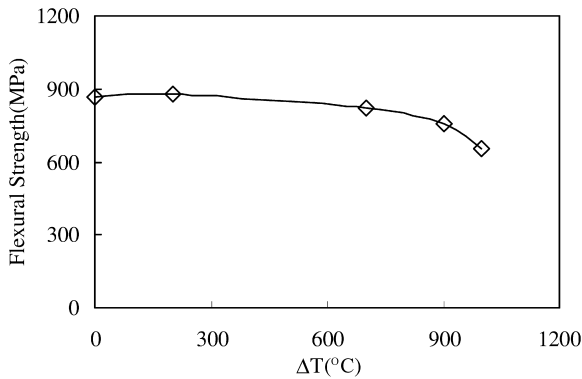


Fig. 6. The effect of quench temperature difference on the flexural strength of C/SiC.

$$\sigma_t = \frac{\alpha E \Delta T_c}{1 - \nu} \quad (1)$$

where E is the Young's modulus, α is the coefficient of thermal expansion, and ν the Poisson's ratio of the material. Regarding Eq. (1), we obtain an expression for ΔT_c

$$\Delta T_c = \frac{\sigma_t(1 - \nu)}{\alpha E}. \quad (2)$$

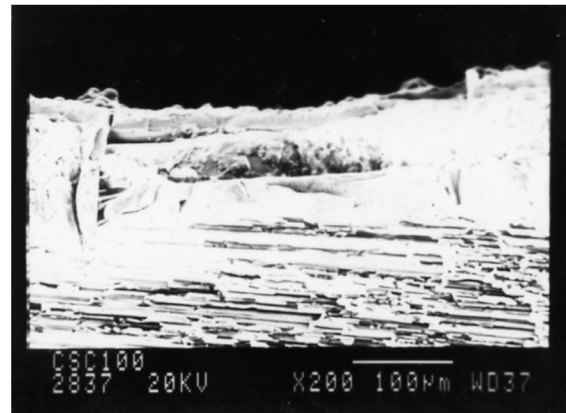
In the present paper, the tensile strength of C/SiC is about 323 MPa, the value of ν is about 0.63, E is about 50 GPa. Therefore, from Eq. (2), $\Delta T_c \cong 683^\circ\text{C}$, which is consistent with the experimental result.

As shown in Fig. 7a, the cracks on the surface of the matrix were formed at lower ΔT , which does not affect the strength. Above ΔT_c (700°C), the propagation of cracks inwards cause the fracture of fibers and the decrease of strength (Fig. 7b). At $\Delta T > 1000^\circ\text{C}$, the interior matrix cracks that propagate through fibers cause a further decrease of strength, as shown in Fig. 7c.

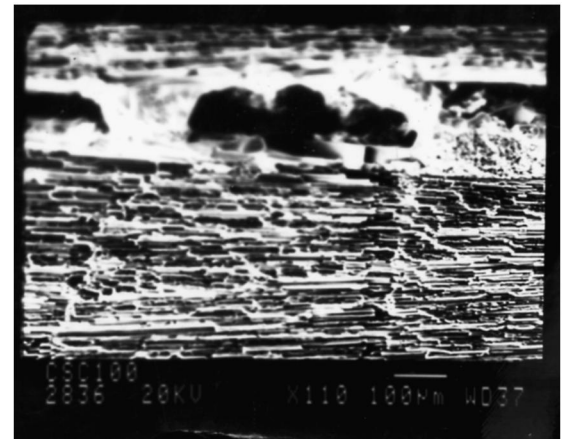
3.2. Effect of quench number on the flexural strength

The driving force for crack propagation is derived from the elastic energy stored in the body caused by repeated cooling and heating. When the elastic energy is larger than the surface energy required for propagation of cracks, the cracks will propagate. During the initial stage of thermal shock, with the increase of quench number, surface cracks propagate inwards, which leads to the increase of crack density in composites. The increase of matrix crack density may result in the decrease of the mechanical properties of composites.

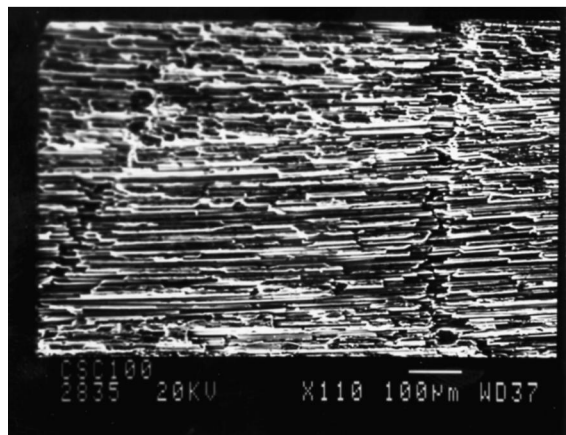
At $\Delta T > \Delta T_c$, the flexural strength as a function of quench cycle number is shown in Fig. 8. As shown in Fig. 8, with the increase of quench cycle number, the flexural strength of the composite decreases gradually. When the



(a)



(b)



(c)

Fig. 7. Morphology of microcracks after thermal shock for 100 cycles. (a) $\Delta T = 600^\circ\text{C}$, (b) $\Delta T = 700^\circ\text{C}$, (c) $\Delta T = 1000^\circ\text{C}$.

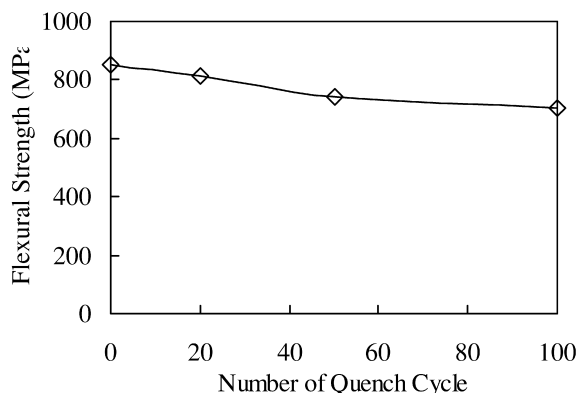
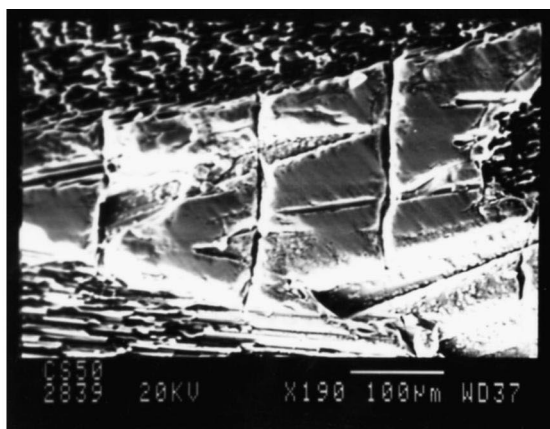
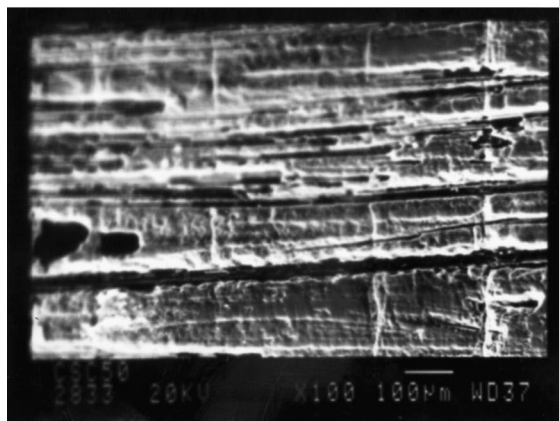


Fig. 8. Effect of thermal-shock times on the flexural strength of C/SiC ($\Delta T=1000^{\circ}\text{C}$).

composite is thermally shocked more than 50 cycles, the strength does not decrease further until the quenching number is as high as 100 cycles. The above results indicate



(a)



(b)

Fig. 9. Morphology of cracks in C/SiC. (a) 50 cycles (b) 100 cycles.

that there exists a critical value for the quench number of C/SiC composites. When the quench number reaches the critical value, the mechanical properties of composites do not decrease with increase of quench number. At the same time, Fig. 8 also indicates that the decrease of mechanical properties of C/SiC during thermal shock results from physical damage, not from chemical corrosion (i.e. oxidation). Otherwise, the strength of composites will continuously decrease with the increase of quench number.

In the present experiment, the critical number of quench cycles is 50 times. As shown in Fig. 9, the crack density is saturated after 50 cycles. With the increase of quench number during the initial stage of thermal shock, cracks propagate into the interior of composites, fiber and matrix debond, and cracks bridge between fibers, which may cause the continuous decrease of strength. With the further increase of quench number, the fibers are damaged, which leads to the observed decrease of strength. When the quenching number is larger than the critical value, the crack density is saturated, and the strength of the composites does not decrease further. After the composite is quenched more than 100 times at $\Delta T=1000^{\circ}\text{C}$, the residual flexural strength is still 83% of the original value.

4. Conclusions

1. C/SiC composites have good thermal shock resistance. After the composite was quenched more than 100 times, the residual flexural strength was 83% of the original value.
2. The ΔT_c of C/SiC is 700°C . Above ΔT_c , the cracks on the surface of the matrix will propagate into the interior of the composites and make the fibers degrade, causing a decrease of the flexural strength.
3. The critical number of thermal shock cycles of C/SiC composite is about 50 cycles. When the number of thermal shocks is less than 50 cycles, the residual flexural strength of C/SiC composite decreases with the increase of thermal shock cycles. When the number of thermal shock cycles of C/SiC is greater than 50, the strength of C/SiC does not further decrease because the crack density is saturated.

Acknowledgements

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